

CS & IT ENGINEERING



Digital Logic

Combinational Circuits

Lecture No. 07



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Recap of Previous Lecture



Topic

MUX

Set 4

Set 5 → 2 HW



Topics to be Covered



Topic

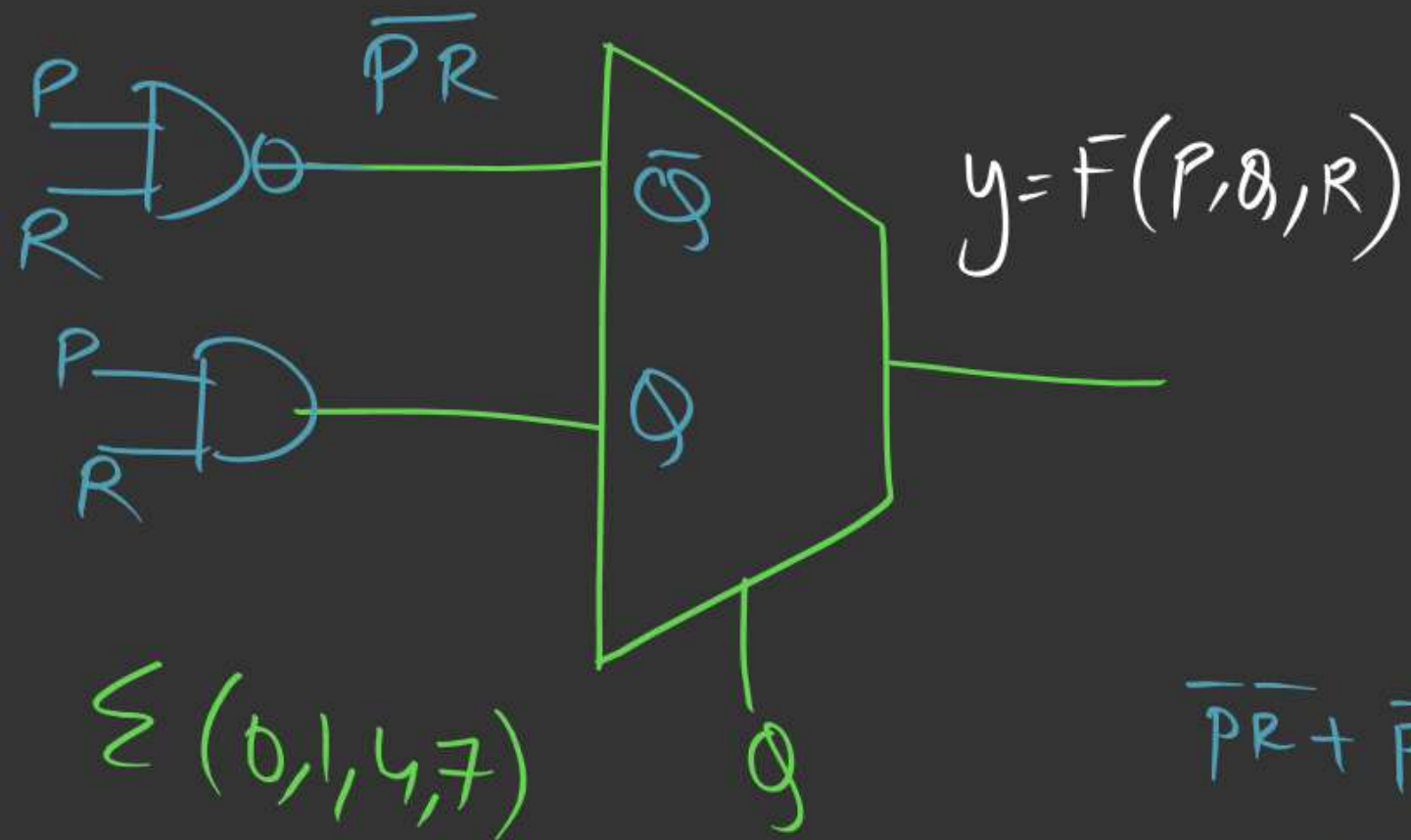


Mux set 6

Demux

Encoder

hw1: 2xMUX, Q as select line.



Standard

	$\bar{Q}$	Q
$\bar{P} \bar{R}$	$\bar{P} \bar{Q} \bar{R}$ (0)	$\bar{P} Q \bar{R}$ (2)
$\bar{P} R$	$\bar{P} \bar{Q} R$ (1)	$\bar{P} Q R$ (3)
$P \bar{R}$	$P \bar{Q} \bar{R}$ (4)	$P Q \bar{R}$ (6)
$P R$	$P \bar{Q} R$ (5)	$P Q R$ (7)

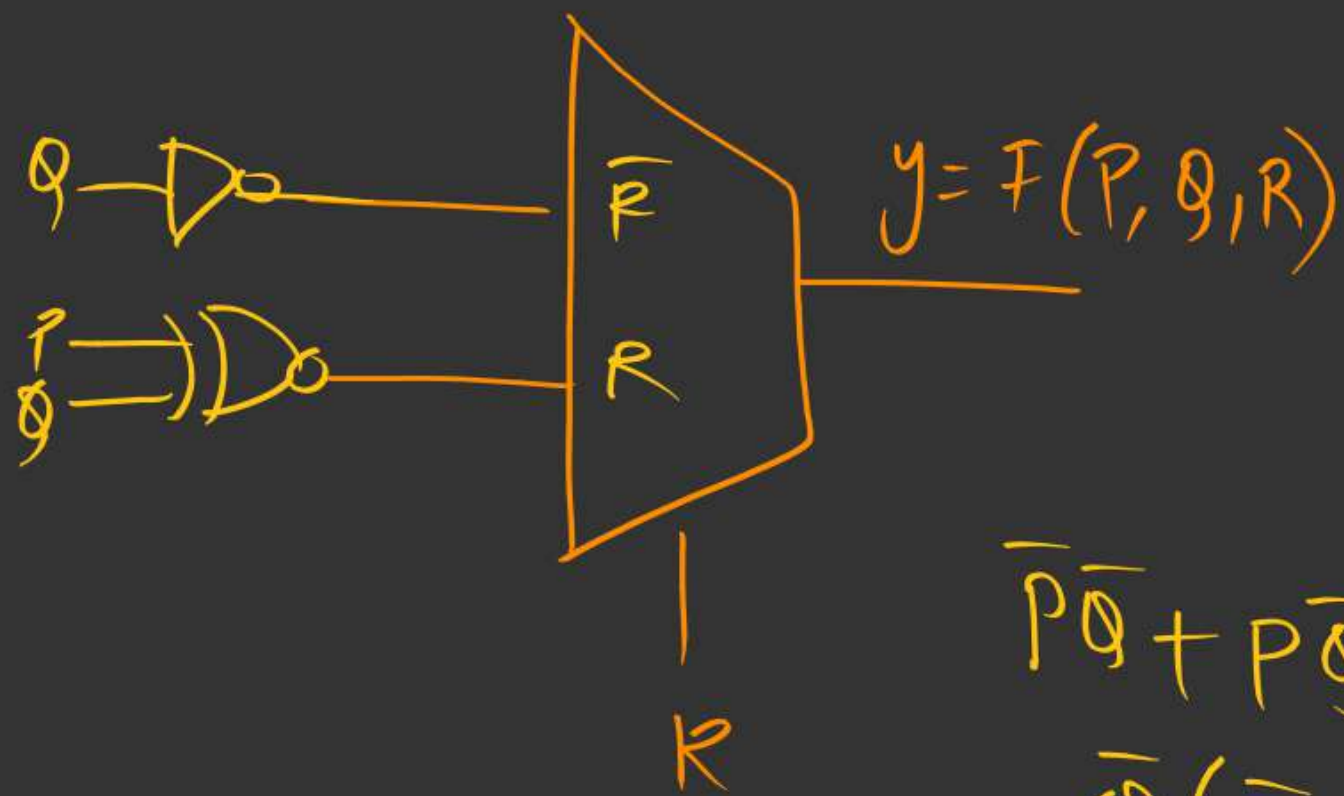
$$\bar{P} \bar{R} + \bar{P} R$$

$$\bar{P} \bar{R} + \bar{P} R = (\bar{P} + R)$$

$P R$

HW2: 2x1 mux, R as select line

$\Sigma(0,1,4,7)$

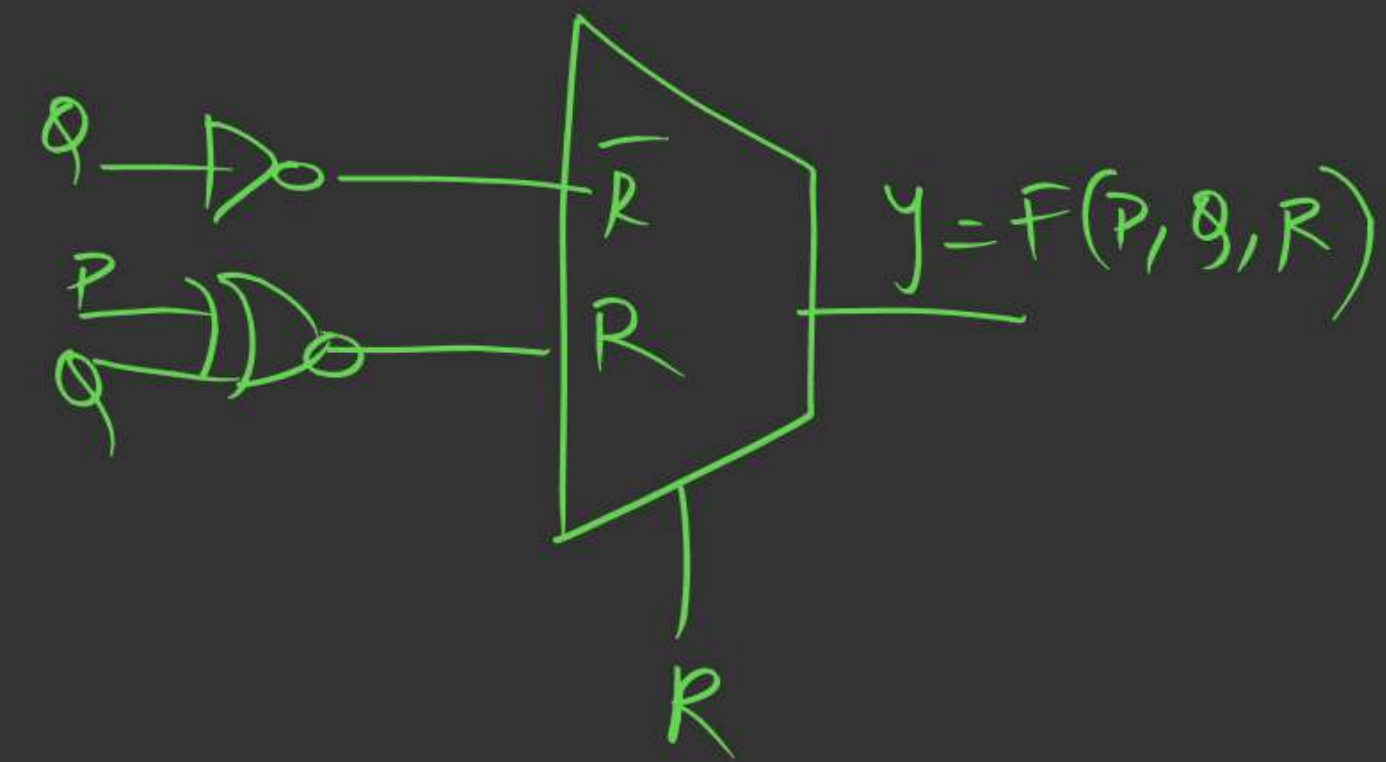


	$\bar{R}$	R
$\bar{P} \bar{Q}$	000 $\bar{P} \bar{Q} \bar{R}$	001 $\bar{P} \bar{Q} R$
$\bar{P} Q$	010 $\bar{P} Q \bar{R}$	011 $\bar{P} Q R$
$P \bar{Q}$	100 $P \bar{Q} \bar{R}$	101 $P \bar{Q} R$
$P Q$	110 $P Q \bar{R}$	111 $P Q R$

$$\bar{P} \bar{Q} + P \bar{Q}$$

$$\bar{Q}(\bar{P} + P) = \bar{Q}$$

$$\bar{P} \bar{Q} + P \bar{Q} = \bar{Q}$$



Short cut $F(P, Q, R) \Sigma(0, 1, 4, 7)$

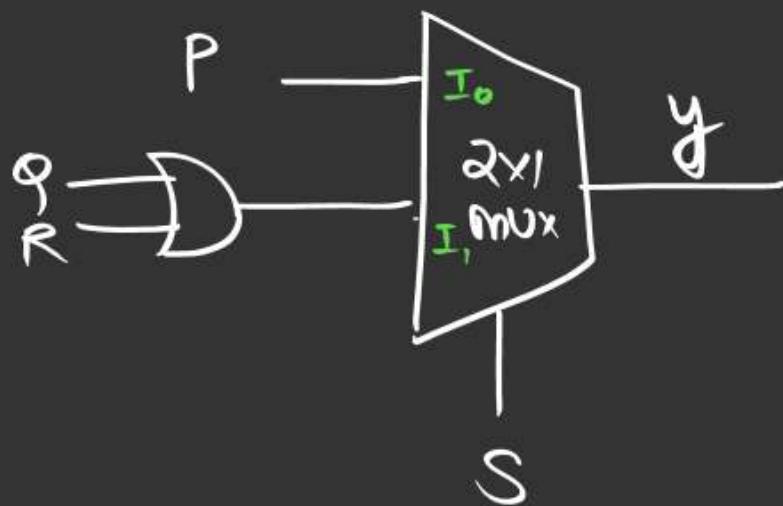
$$= \bar{P}\bar{Q}\bar{R} + \bar{P}\bar{Q}R + P\bar{Q}\bar{R} + PQ\bar{R}$$

$$= (\bar{P}\bar{Q})\bar{R} + (\bar{P}\bar{Q})R + (P\bar{Q})\bar{R} + (PQ)\bar{R}$$

$$= (\bar{P}\bar{Q} + P\bar{Q})\bar{R} + (\bar{P}\bar{Q} + PQ)R$$

$$= (\bar{Q})\bar{R} + (P \odot Q)R$$

Set 6 : MUX with delay



3
5

$$d_{OR} = 2 \mu s$$

$$d_{mux} = 3 \mu s$$

Case 1 : If $S = 0$, $O/P = I_0$

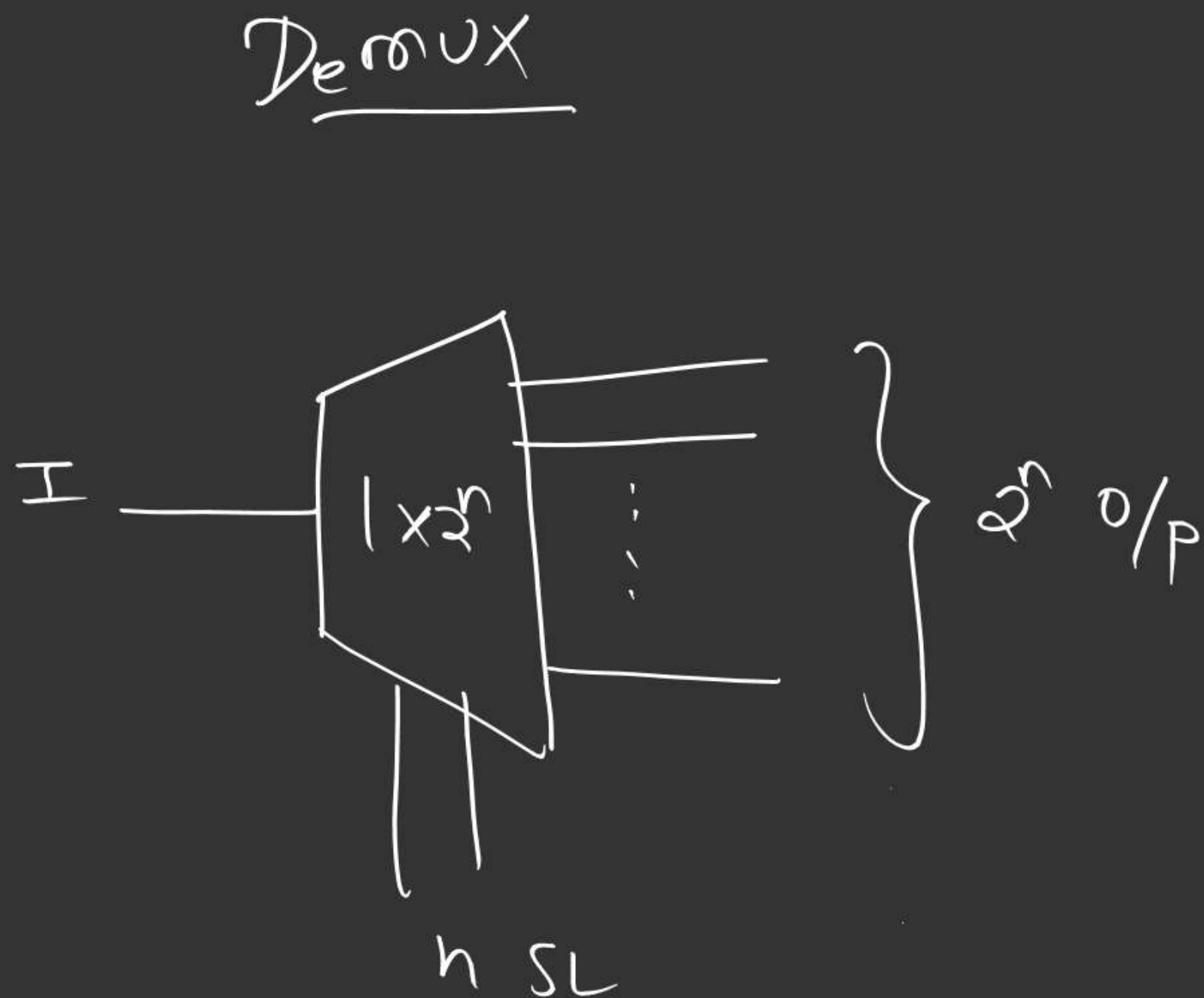
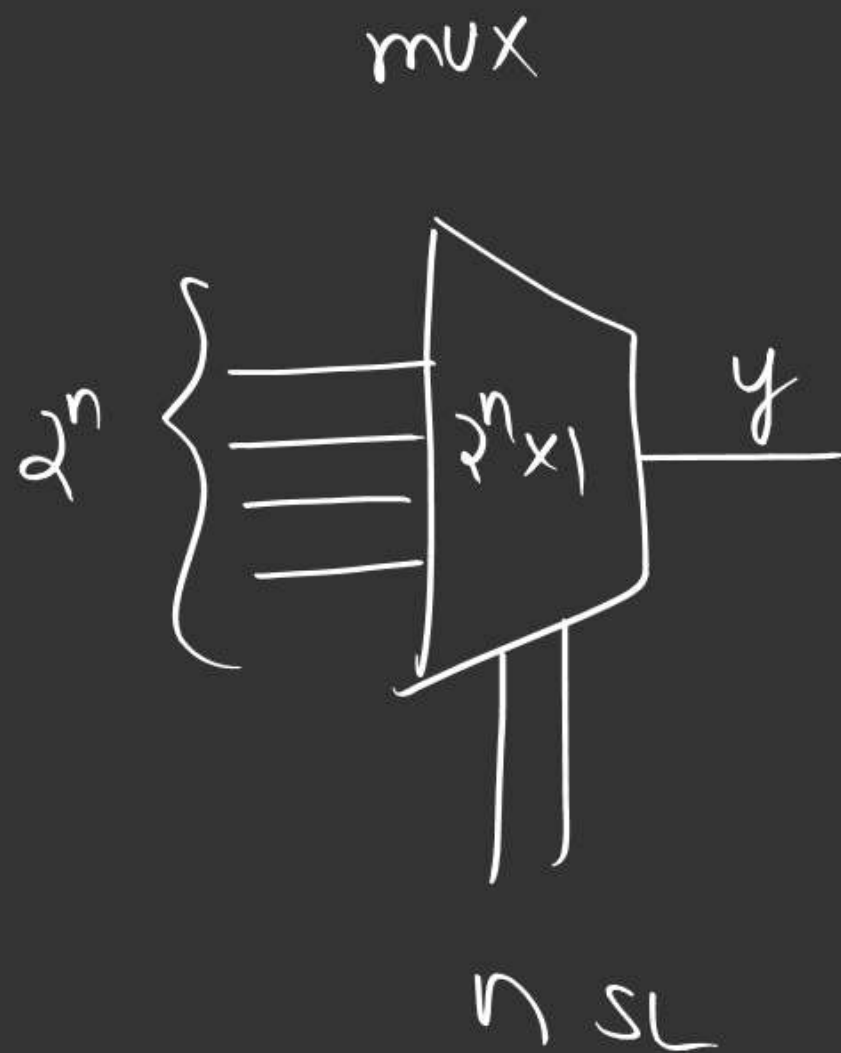
$$\text{delay} = d_{mux} = 3 \mu s$$

Case 2 : If $S = 1$, $O/P = I_1$

$$\begin{aligned} \text{delay} &= d_{OR} + d_{mux} \\ &= 2 + 3 = 5 \mu s \end{aligned}$$

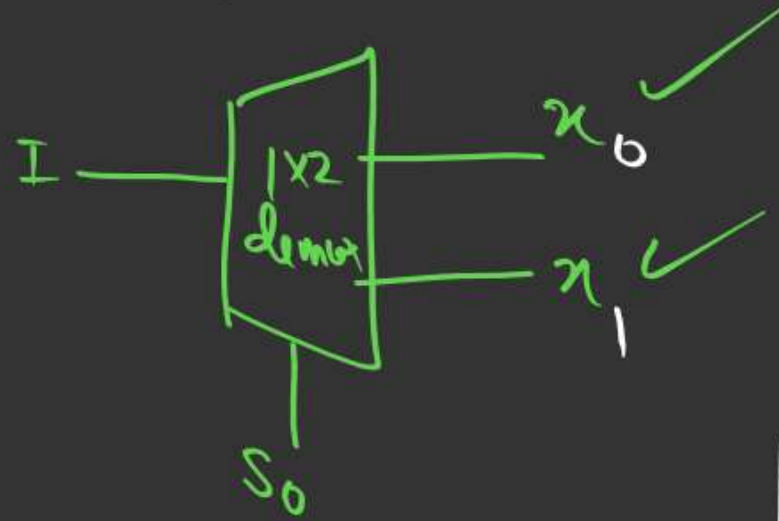


→ mux completed



(Q.1) Design a 1x2 dm mux.

Step 1:- no. of i/p & o/p,



Step 2:- Truth table.

S ₀	x ₀	x ₁
0	I	0
1	0	I

Step 3:- Logical expression.

$$x_0 = \overline{S_0} \cdot I$$

$$x_1 = S_0 \cdot I$$

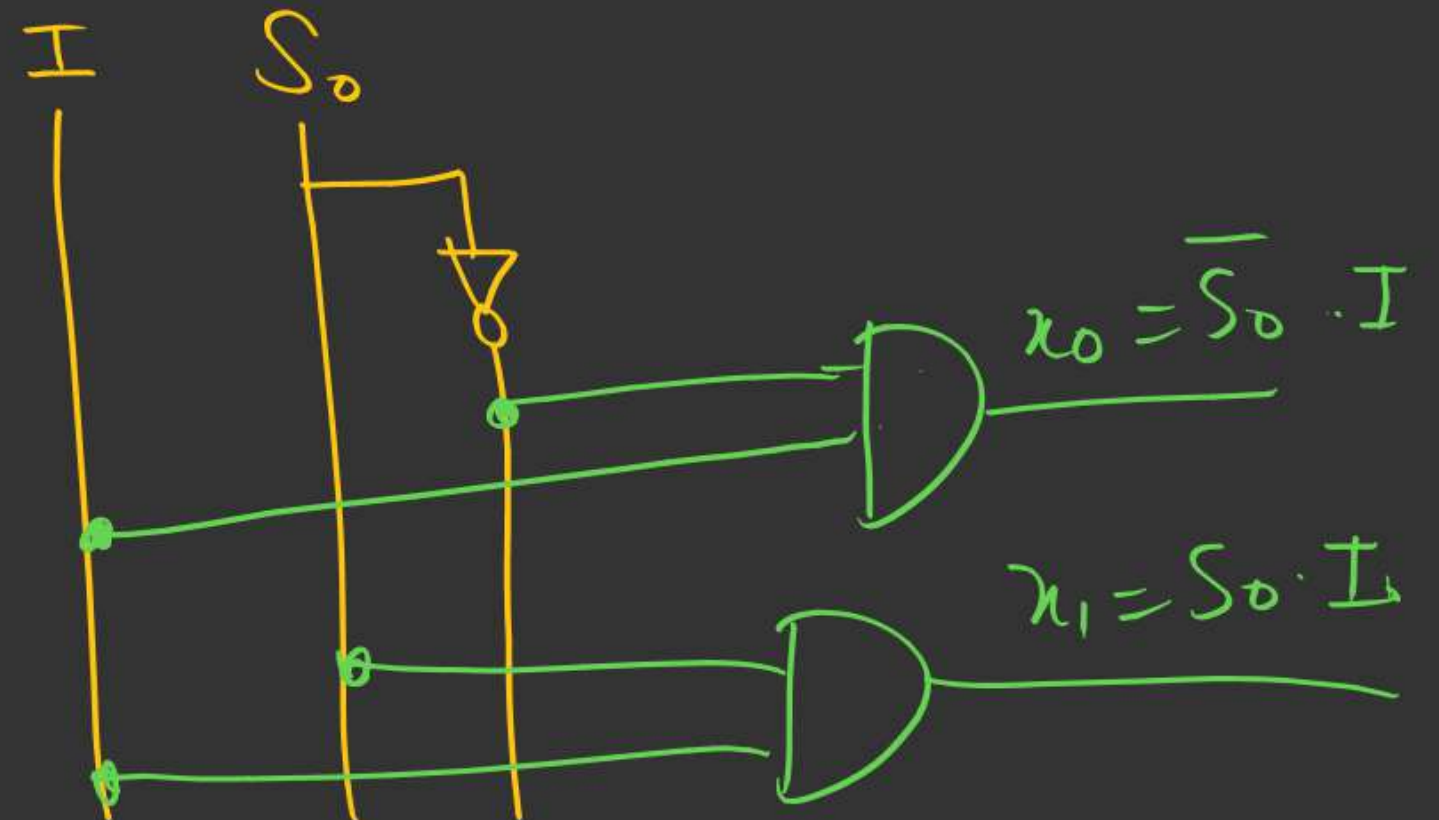
Step 4 :- minimize

$$x_0 = \overline{S_0} \cdot I$$

$$x_1 = S_0 \cdot I$$

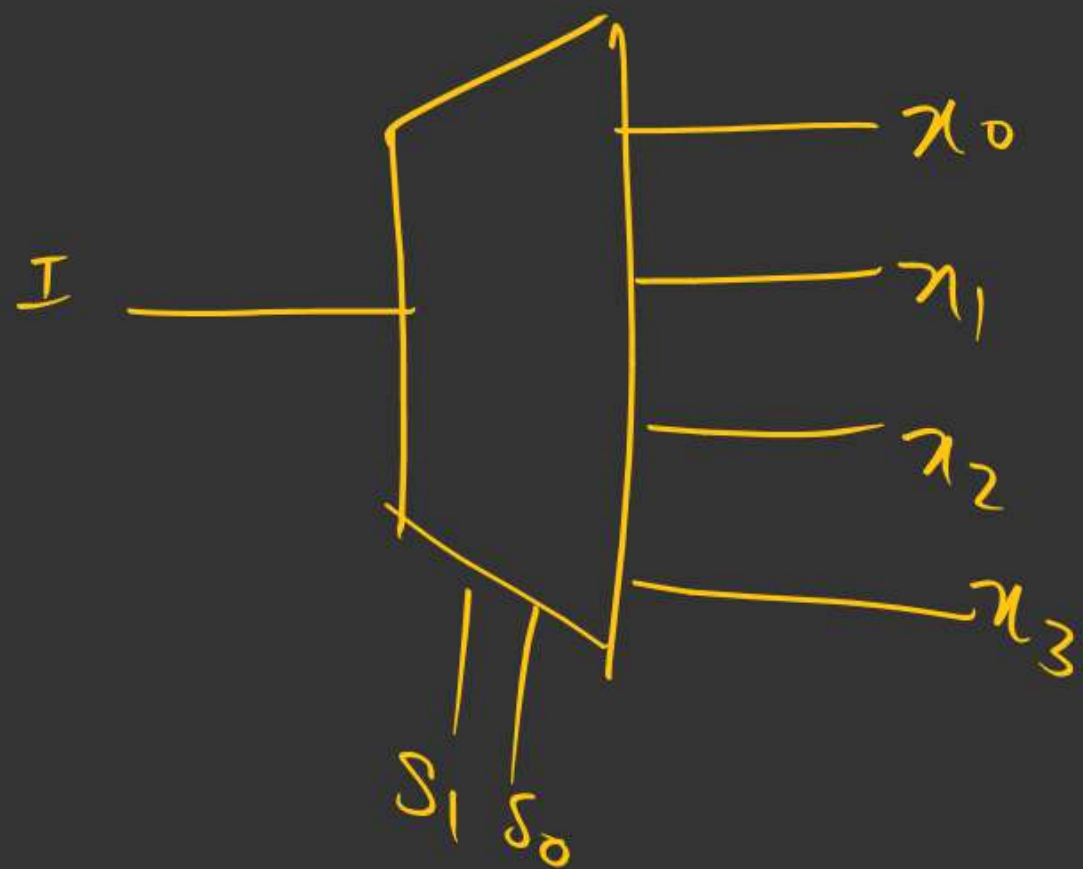
already minimized

Step 5: Design Hardware.



(Q.2) Design 1X4 de-mux

1) i/p & o/p



2) Truth table

	S_1	S_0	x_0	x_1	x_2	x_3
	0	0	I	0	0	0
	0	1	0	I	0	0
	1	0	0	0	I	0
	1	1	0	0	0	I

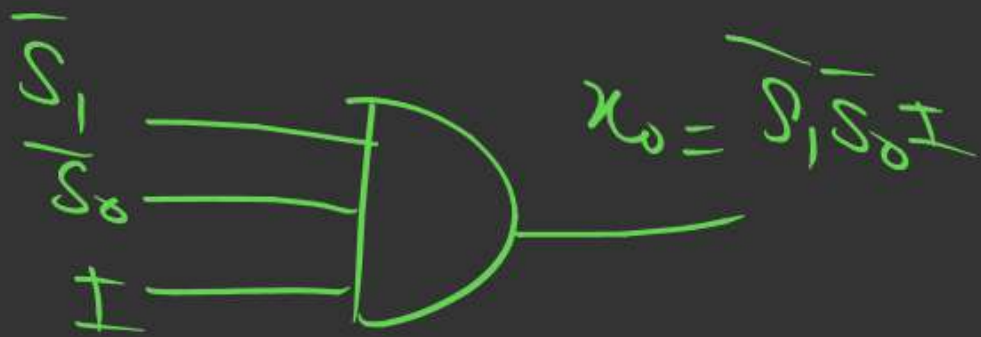
$$3) \quad x_0 = \bar{S}_1 \bar{S}_0 I$$

$$x_1 = \bar{S}_1 S_0 I$$

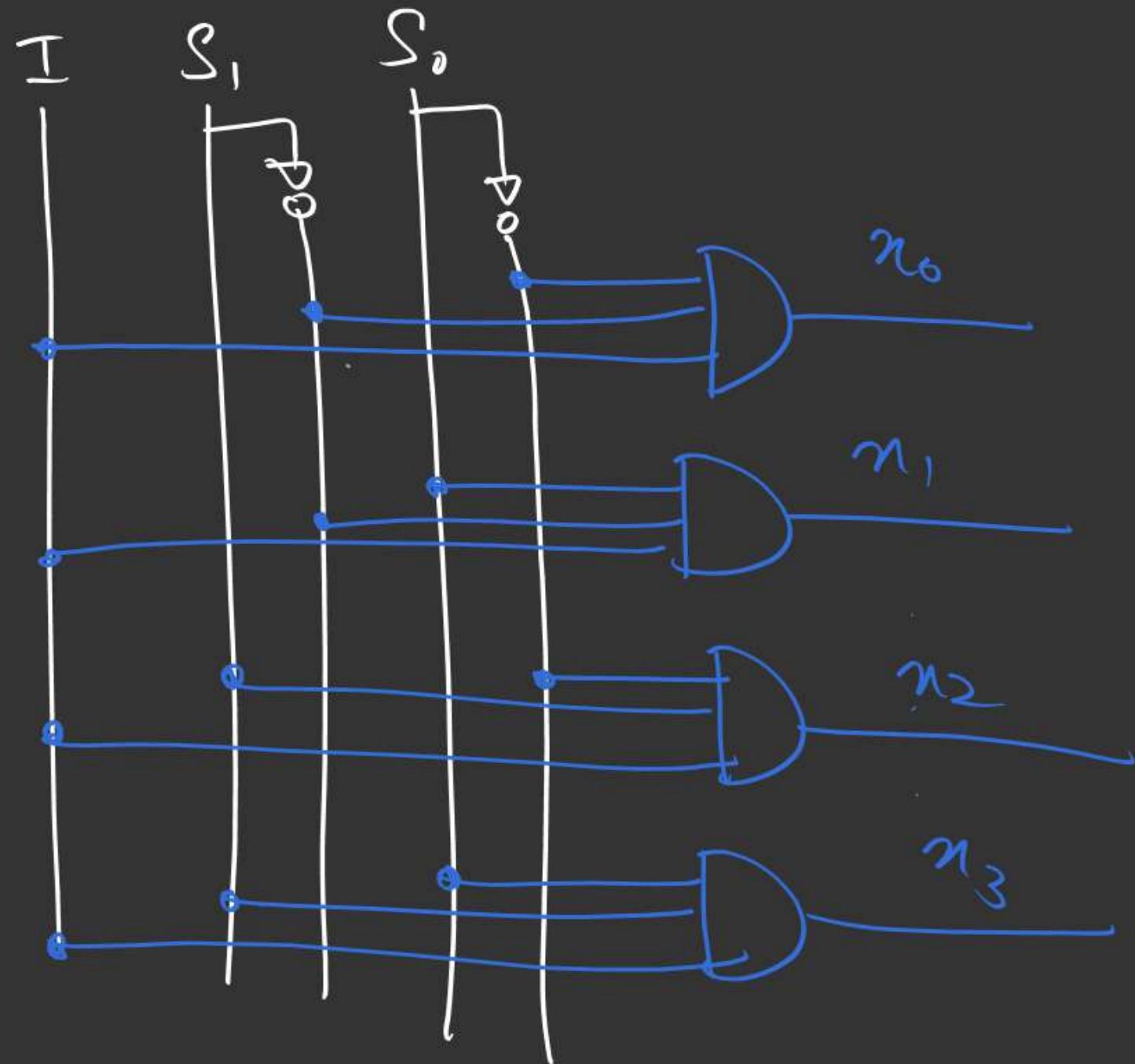
$$x_2 = S_1 \bar{S}_0 I$$

$$x_3 = S_1 S_0 I$$

4) minimize ↗



5) Diagram



HW

1) Follow 5 steps for 1×8 de-mux

2) " " " 1×16 de-mux

★ Encoder

eg

A	→	1
B	→	5
C	→	7
D	→	2



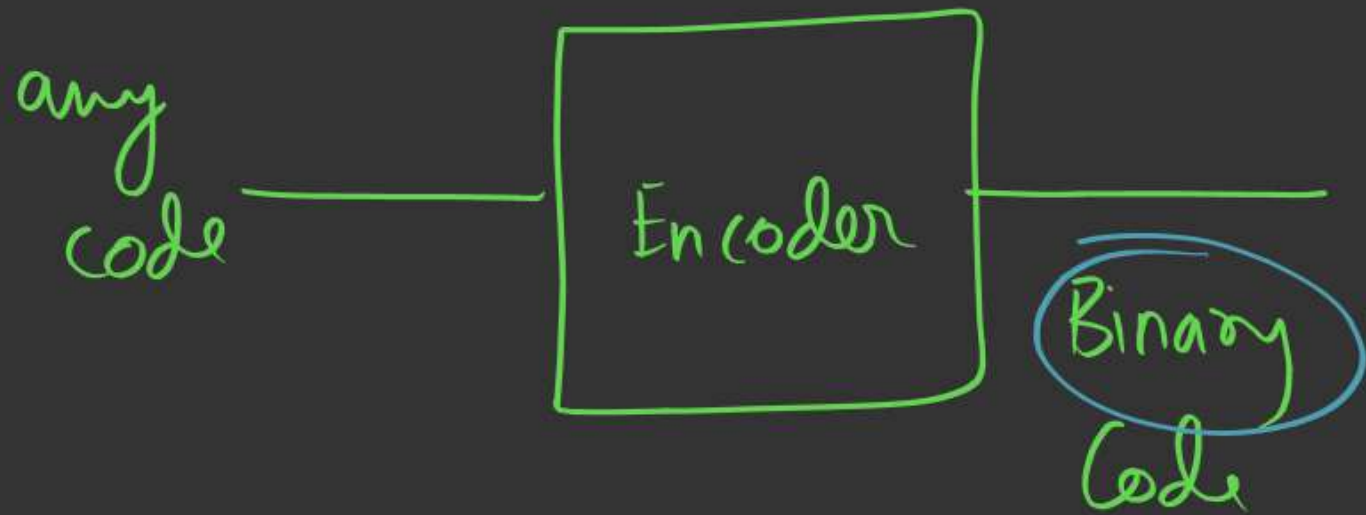
ACD → (172)

→ A circuit that converts any given code
into Binary Code.

2 bits

A	→	00
B	→	01
C	→	10
D	→	11

$$2^n \times n$$



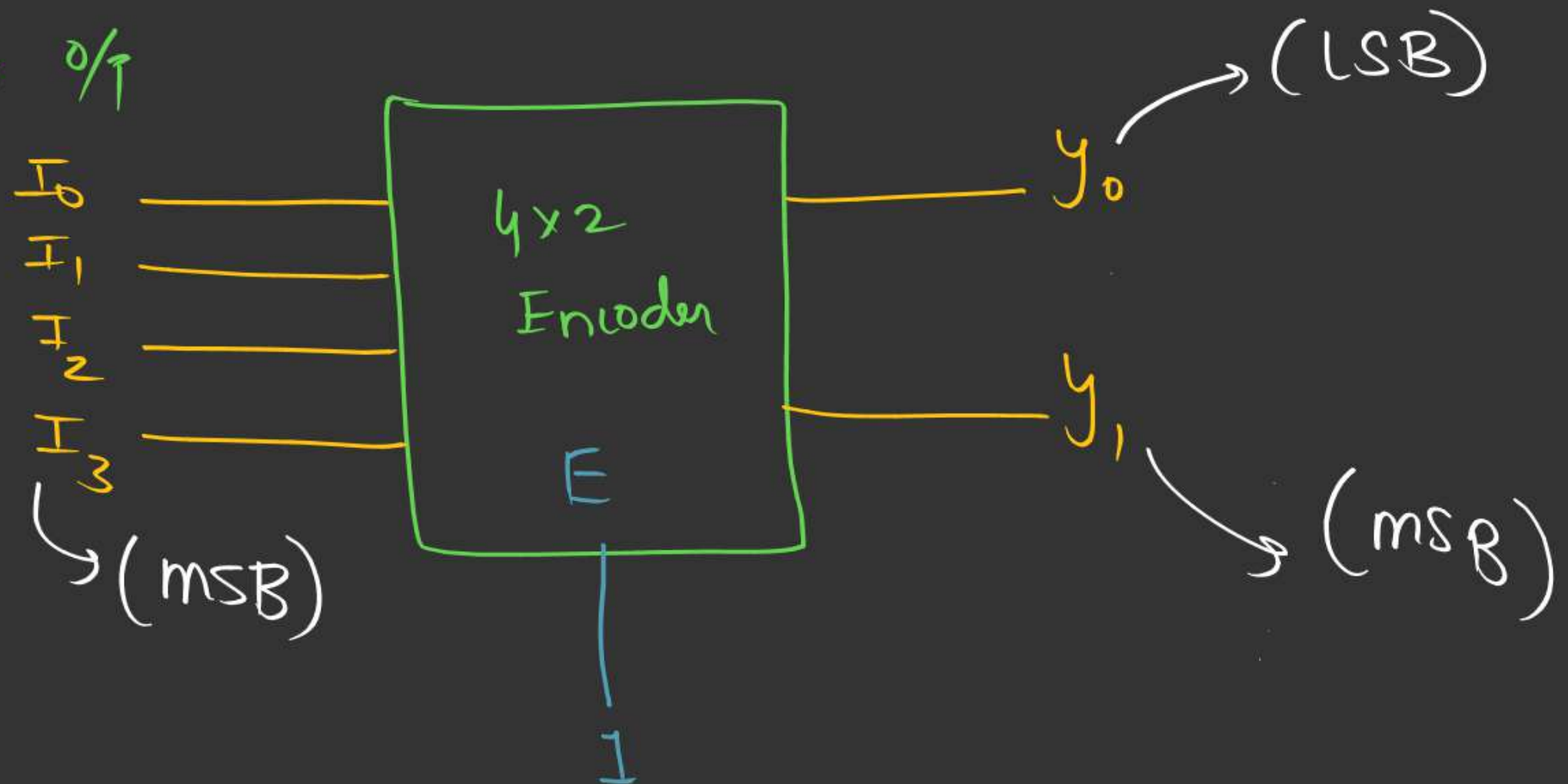
- eg:-
- 1) 4×2 Encoder (Quad to Binary)
 - 2) 8×3 Encoder (Octa to Binary)
 - 3) 16×4 Encoder (Hexa to Binary)

> Design 4x2 Encoder. ^(default) (Standard ordering)

Step 1: - ^{1/p} & ^{0/1}
(LSB)

Enable = 1 (ON)

= 0 (OFF)



2) Truth Table

i/p				o/p	
I_3	I_2	I_1	I_0	y_1	y_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
✓1	0	0	0	1	1

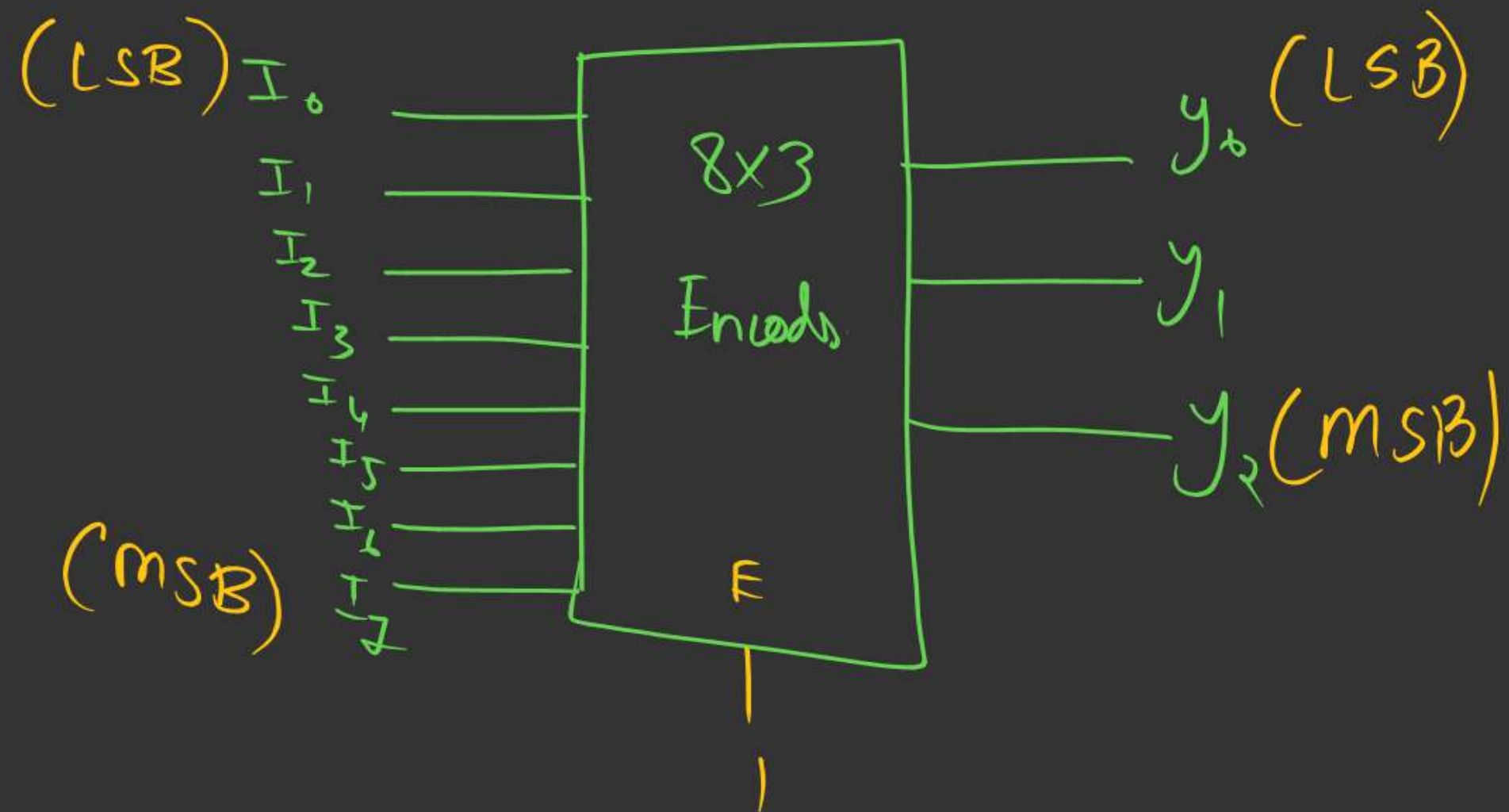
Step 3:- logical expression.

$$y_0 = I_1 + I_3$$

$$y_1 = I_2 + I_3$$

(8) Design 8x3 Encoder (similarly)

Step 1: i/p & o/p



Step 2:- Tenth table :-

i/p								o/p		
MSB								MSB	LSB	
I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	y_2	y_1	y_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

Step 3: Logical expression:

$$y_0 = I_1 + I_3 + I_5 + I_7$$

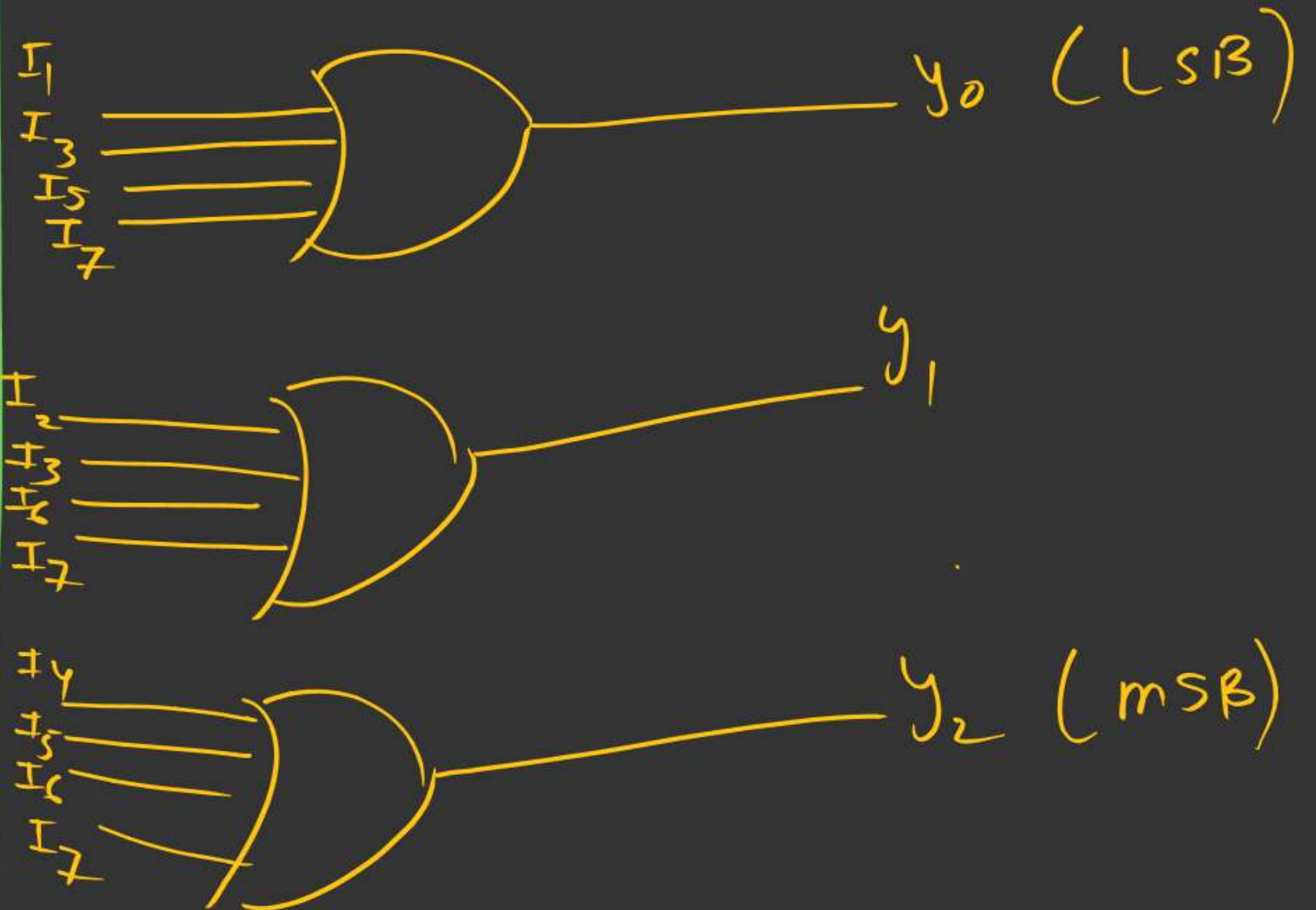
$$y_1 = I_2 + I_3 + I_6 + I_7$$

$$y_2 = I_4 + I_5 + I_6 + I_7$$

Step 4: Minimize:

is already done.

Step 5:-



HW

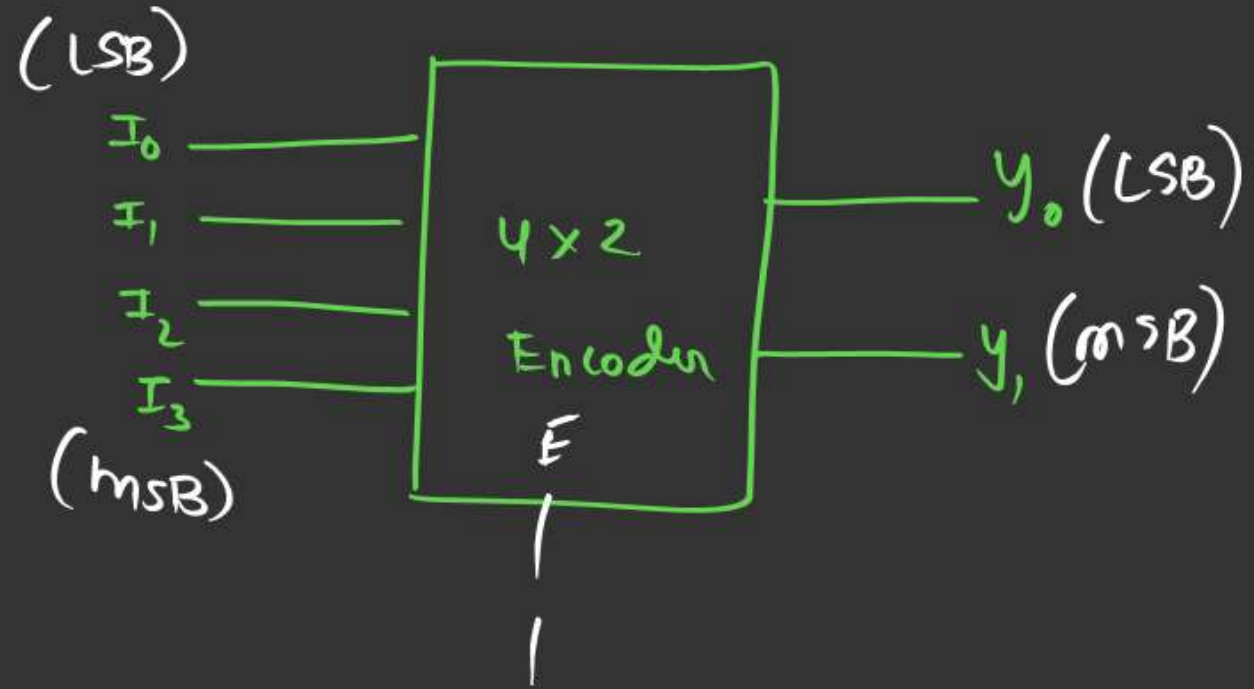
Design

16X4

Encoder (

framework)

eg:- Priority Encoder



HW

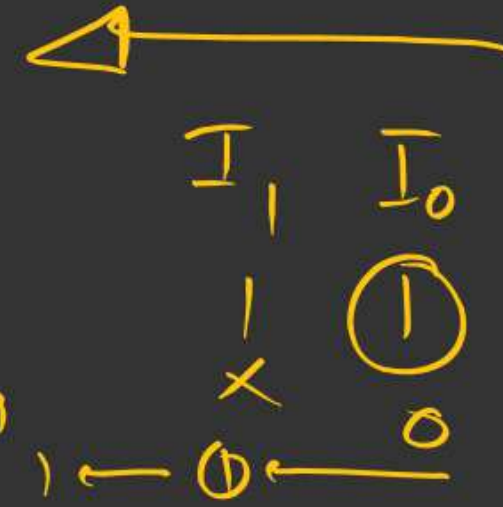
→ try msb Pri Enc

LSB Priority Encoder.

Truth Table

I_3	I_2	I_1	I_0	y_1	y_0
X	X	X	1	0	0
X	X	1	0	0	1
X	1	0	0	1	0
1	0	0	0	1	1

Priority Encoder



1) LSB Priority Encoder → LSB ✓

2) MSB Priority Encoder → MSB





2 mins Summary



→ MUX → set 5 ✓
 └→ set 6 ✓

→ De-MUX design

→ Encoder

→ Priority Encoder





THANK - YOU

